

MANZANILLO, Mexico

WMO: 766540

Lat: 19.0444N

Lon: 104.3186W

Elev: 10

StdP: 14.69

Time Zone: -6.00 (MEX)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
3	65.3	66.9	51.5	56.6	72.2	54.8	64.0	72.9	14.8	77.3	13.4	77.8	3.4	0	0.293

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.7	92.5	79.6	91.5	79.8	90.6	79.7	83.1	89.2	82.3	88.8	81.6	88.2	5.5	230

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
81.4	163.2	87.6	80.5	158.4	87.2	79.8	154.5	86.6	47.0	89.6	46.1	89.0	45.3	88.3	90.5

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 13.8	11.3	9.6	60.3	94.3	3.5	3.4	57.8	96.7	55.8	98.7	53.9	100.6	51.3	103.0		
(5)			54.8	84.8	2.7	1.9	52.9	86.1	51.3	87.2	49.7	88.3	47.8	89.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6)	Temperatures, Degree-Days and Degree-Hours	DBAvg	80.4	77.2	76.7	76.1	76.6	80.1	83.7	84.1	84.0	83.3	83.0	81.3	78.6	(6)
(7)		DBStd	3.89	2.57	2.56	2.92	2.66	3.01	2.27	2.09	1.92	2.01	1.94	2.15	2.19	(7)
(8)		HDD50	0	0	0	0	0	0	0	0	0	0	0	0	0	(8)
(9)		HDD65	0	0	0	0	0	0	0	0	0	0	0	0	0	(9)
(10)		CDD50	11099	843	749	810	797	932	1012	1057	1053	998	1024	939	885	(10)
(11)		CDD65	5624	378	329	345	347	467	562	592	588	548	559	489	420	(11)
(12)		CDH74	56531	2665	2288	2346	2491	4549	6797	7368	7131	6354	6429	4877	3236	(12)
(13)		CDH80	19584	640	496	503	518	1354	2748	3145	2942	2410	2385	1589	854	(13)
(14)	Wind	WSAvg	4.4	4.2	4.2	4.5	4.6	4.7	4.6	4.7	4.7	4.1	4.0	4.1	(14)	
(15)	Precipitation	PrecAvg	39.0	0.4	0.5	0.0	0.0	0.0	4.3	6.8	8.3	10.7	5.1	0.8	0.6	(15)
(16)		PrecMax	60.2	4.7	7.8	0.4	0.2	0.7	13.1	23.2	28.0	30.2	23.1	11.0	3.7	(16)
(17)		PrecMin	24.7	0.0	0.0	0.0	0.0	0.0	0.1	0.6	0.5	1.3	0.0	0.0	0.0	(17)
(18)		PrecStd	10.7	0.9	1.5	0.1	0.0	0.2	3.7	5.3	5.7	6.3	4.7	2.0	1.0	(18)
(19)	Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	88.2	87.7	88.2	88.7	90.8	93.5	93.6	93.9	92.5	91.8	90.7	88.5	(19)
(20)			MCWB	72.8	75.0	73.2	74.0	78.0	78.2	80.4	78.7	79.9	80.5	78.1	76.0	(20)
(21)		2%	DB	86.5	86.0	86.2	86.4	89.0	91.8	92.2	92.2	90.9	90.7	89.2	87.2	(21)
(22)			MCWB	73.7	75.2	73.3	74.7	78.5	78.7	80.4	79.9	80.1	80.4	78.3	75.8	(22)
(23)		5%	DB	85.1	84.6	84.6	84.8	87.4	90.4	91.0	91.0	89.8	89.4	87.9	86.0	(23)
(24)			MCWB	73.9	74.7	73.6	74.9	78.1	78.9	80.1	80.1	80.1	80.3	77.8	75.4	(24)
(25)		10%	DB	83.7	83.2	83.0	83.1	85.9	88.9	89.7	89.7	88.6	88.2	86.7	84.7	(25)
(26)			MCWB	73.9	74.1	73.3	74.1	77.3	78.8	79.8	79.9	79.9	79.9	77.6	75.1	(26)
(27)	Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	78.3	78.3	78.1	79.5	82.8	84.2	83.7	83.5	83.4	83.6	81.9	79.5	(27)
(28)			MCDWB	84.3	84.6	84.8	85.4	87.6	88.8	90.4	91.0	89.0	89.1	88.3	85.7	(28)
(29)		2%	WB	77.0	77.1	76.5	77.7	80.9	82.5	82.6	82.3	82.2	82.5	80.8	78.4	(29)
(30)			MCDWB	83.4	84.0	83.7	84.2	86.3	87.9	89.7	89.8	88.6	88.4	86.9	84.7	(30)
(31)		5%	WB	76.0	76.1	75.4	76.4	79.2	81.2	81.6	81.5	81.4	81.7	79.8	77.4	(31)
(32)			MCDWB	82.7	83.0	82.6	82.9	85.4	87.3	88.8	88.9	87.8	87.6	85.9	83.8	(32)
(33)		10%	WB	75.1	75.1	74.3	75.1	78.1	80.1	80.7	80.7	80.7	80.8	78.9	76.4	(33)
(34)			MCDWB	81.9	81.7	81.3	81.6	84.5	86.6	88.0	88.1	87.1	86.8	84.9	82.8	(34)
(35)	Mean Daily Temperature Range	5% DB	MDBR	13.6	13.3	13.7	13.3	11.1	9.9	10.7	11.1	10.3	10.5	11.7	12.7	(35)
(36)			MCDBR	14.2	14.0	14.6	14.5	11.8	11.6	12.2	12.7	11.9	11.5	12.0	13.4	(36)
(37)			MCWBR	9.5	9.2	9.2	9.1	6.7	5.6	6.3	6.8	5.9	6.0	6.4	8.4	(37)
(38)		5% WB	MCDBR	13.2	13.1	13.6	13.5	11.2	9.9	11.3	11.9	11.0	10.6	10.9	12.2	(38)
(39)			MCWBR	9.6	9.5	9.3	9.4	7.5	5.8	6.6	7.0	6.1	6.3	6.6	8.1	(39)
(40)	Clear-Sky Solar Irradiance	taub	0.386	0.386	0.376	0.425	0.530	0.511	0.477	0.467	0.464	0.423	0.386	0.388	(40)	
(41)		taud	2.450	2.424	2.423	2.277	2.056	2.155	2.266	2.292	2.300	2.424	2.505	2.472	(41)	
(42)		Ebn at Noon	283	290	296	280	249	252	260	265	266	274	281	279	(42)	
(43)		Edh at Noon	34	36	37	44	54	49	43	43	42	36	32	32	(43)	
(44)	All-Sky Solar Radiation	RadAvg	1543	1841	2189	2357	2253	1916	1831	1794	1624	1638	1588	1476	(44)	
(45)		RadStd	108	157	92	74	121	108	123	87	111	108	122	76	(45)	

Historical Trends

	DBAvg	Heating			Cooling			Degree-Days			
		99% DB		99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (1 neighbor)	+0.97	+2.54	-3.29	N/A	N/A	N/A	N/A	N/A	-193	+410	+234

Nomenclature: See separate page